

Lullaby

Product Requirements Document

“A story doesn’t have one correct form. Lullaby makes that literal.”

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TIMELINE 36 hours

TRACK
Entertainment &
Media

1. Product Summary

Lullaby is a web application that turns any spoken story into a fully narrated animated film. The user speaks their story aloud, and Lullaby generates a complete illustrated film with AI narration in under 90 seconds.

Before recording, the user configures the film across five independent filter axes: visual style, narrator voice, reading level, tone, and pacing. The same thirty-second story can produce a toddler-level bedtime tale in a grandmother’s voice with watercolor illustrations, or a literary adventure in a theatrical voice with Ghibli-style animation — from identical input.

The user’s own voice is not used as audio. The AI handles narration, visuals, and ambient sound. The user is the origin of the story, not its vessel. The filter system encodes a specific belief: there is no single correct form of a story. Every story exists in many simultaneous versions, each one complete, each one serving a different person at a different moment. Lullaby makes that literal.

The filter matrix produces 2,160 unique film configurations from a single spoken input — not as cosmetic overlays on the same asset, but as genuinely different artifacts that serve different children, different ages, different moods, different nights.

Who It’s For

- Parents and grandparents telling bedtime stories to children ages 2 through 12
- Teachers and librarians producing quick read-alouds for different age groups
- Writers and hobbyists who want to hear and see their ideas performed
- Anyone preserving a family story, a memory, or a dream as a shareable artifact

2. Goals and Non-Goals

Goals

- Generate a fully narrated illustrated film from spoken input in under 90 seconds
- Support five independent filter axes the user configures before speaking: visual style, narrator voice, reading level, tone, and pacing
- Produce filter swaps that regenerate only affected layers, keeping refilter time under 30 seconds
- Silently clean transcript artifacts (pauses, filler words, obvious self-corrections) so they do not appear in the film
- Maintain visual consistency of characters and settings across all scenes in a film
- Use only the video generation stack's own audio capabilities — no external TTS services
- Ship a working end-to-end product in 36 hours on free tiers

Non-Goals

- Using the speaker's own voice as narration audio — narration is always AI-generated
- External TTS services (ElevenLabs, Coqui, Play.ht) — audio must come from the video generation pipeline itself
- Advanced narrative reconstruction from deeply fragmented input — the system cleans artifacts, it does not rewrite broken narratives
- Voice cloning of real individuals
- Interactive or branching stories
- Post-generation editing of the finished film
- Stories longer than 5 minutes in the hackathon build

3. User Flow

The experience is a single uninterrupted ritual from decision to playback. No menus, settings pages, or configuration steps break the flow. The interaction is designed to feel like a ceremony — the kind of deliberate act that makes a story feel worth telling.

Step 1 — Landing

Dark screen. One softly glowing crescent moon. One line: “Tell a story. Watch it become a film.” One button: Begin.

Step 2 — The Picker

A single elegant picker screen with five filter axes. Each axis shows its current value; tapping reveals alternatives. Sensible defaults are pre-selected (Watercolor, Warm Mother narrator, Early Reader level, Cozy tone, Unhurried pacing), so users can skip to recording or customize any axis they care about.

Step 3 — Record

A soft pulsing candle flame. Microphone indicator. No timer. User tells their story. Natural pauses, occasional filler words, and minor self-corrections are permitted and will be cleaned

silently. When the user stops speaking for 4 seconds, the candle expands, a soft chime plays, and the screen transitions.

Step 4 — The Weaving Screen

A purpose-built animated screen that hides backend latency — and makes the product’s core thesis visible. The cleaned story appears as handwritten calligraphy flowing upward, word by word, at the approximate rhythm of the user’s original speech (timestamps from Whisper). Then, as the rewrite runs, the words slowly reform: the user’s phrasing dissolves and the narrator’s phrasing takes its place. Character sketches appear on screen edges. The main canvas paints scenes in the chosen visual style. A timeline at the bottom fills with gold circles as each scene completes.

The transformation of words on screen — from the user’s voice into the narrator’s voice — is not decorative. It is the story of what the product does, made visible. No spinners, no percentage bars, no status text.

Step 5 — Premiere

All gold circles filled. Play button fades in. Film plays full-screen with AI narration in the chosen voice, generated illustrations synced to narration timing, and ambient sound effects beneath.

Step 6 — Refilter

After the film ends, a prompt: “Try a different version.” The picker returns. Changing any axis regenerates only the affected pipeline layers while keeping expensive work cached. Changing reading level re-runs the rewrite and narration; changing visual style re-runs images; changing narrator voice re-runs audio only.

Step 7 — Save or Restart

Two buttons: “Save as video” (downloads MP4) and “Tell another story” (returns to Step 1).

4. The Filter System

Five orthogonal axes. User picks one option per axis. $6 \times 6 \times 4 \times 5 \times 3 = 2,160$ unique film configurations from a single spoken input.

The filter matrix is the product’s primary differentiator, but not because of the number. Because of what the number implies: a story told once does not have one correct form. The 2,160 configurations are not variations of the same film — they are 2,160 distinct films, each one coherent, each one serving someone specific. A three-year-old and an eleven-year-old can hear the same story their grandmother told, each version true to the plot, each voice right for the moment.

4.1 Visual Style (6 options)

Determines the artistic rendering of all generated imagery. Each style has its own prompt prefix, negative prompt, and per-filter scene-appearance animation.

— **Watercolor Storybook** — Watercolor Storybook

Soft painterly children's-book aesthetic, Beatrix Potter lineage

- **Pixar / 3D Animation** — Pixar / 3D Animation

Rendered, vibrant, cinematic family-film look

- **Studio Ghibli** — Studio Ghibli

Hand-painted anime, warm palette, Miyazaki-inspired backgrounds

- **Paper Cutout** — Paper Cutout

Layered construction-paper stop-motion aesthetic

- **Charcoal Sketch** — Charcoal Sketch

Black and white, moody, dreamlike

- **Crayon Drawing** — Crayon Drawing

Child's art style, wobbly lines, joyful imperfection

4.2 Narrator Voice (6 options)

All voices are generated by the video stack's native audio generation — no external TTS service. The no-external-TTS constraint is not a technical limitation; it is a product philosophy.

Narration and imagery share the same origin model. The film's voice is native to the generation layer. Coherence is by design.

- Warm Mother — soft, maternal, gentle pacing (female, mid-adult)
- Wise Grandfather — aged, slightly gravelly, storyteller cadence (male, older)
- Playful Sister — young, energetic, higher pitch, warm (female, young-adult)
- Gentle Father — calm, steady, reassuring baritone (male, mid-adult)
- Mysterious Narrator — theatrical, slightly neutral gender, suited to fairy tales
- Kid Narrator — a child's voice, as if a seven-year-old is telling the story

4.3 Reading Level (4 options)

The rewrite model transforms the story into the chosen reading level before narration.

Vocabulary, sentence length, and complexity shift meaningfully at each level; plot and characters are preserved exactly. This is the filter that delivers the most tangible user value: a parent can speak a story once and replay it at two different levels for two different children.

- Toddler (2–3) — very short sentences, simple high-frequency vocabulary, repetition, concrete sensory detail
- Early Reader (4–6) — simple but varied sentences, clear cause-and-effect, expressive but age-appropriate
- Grade School (7–10) — richer vocabulary, longer sentences, nuanced emotions, light metaphor
- Advanced (11+) — literary, descriptive, poetic, complex sentence structure — suitable for older kids or adult listeners

4.4 Tone (5 options)

Tone shapes three things simultaneously: how the model rewrites the story, how the narrator performs it, and the visual mood of generated imagery.

- Cozy — warm, gentle, safe, bedtime-appropriate (default)
- Adventurous — energetic, forward-moving, emphasizes action

- Whimsical — playful, leans into surprise and absurd detail
- Mysterious — suspenseful, atmospheric, lingers on unknowns
- Tender — quiet, emotional, pauses on feelings

4.5 Pacing (3 options)

Controls narration speed and per-scene shot duration. Slower pacing produces longer-lingering visuals; faster pacing produces shorter, snappier shots.

- Unhurried — long pauses, slow narration, lingering shots (bedtime wind-down)
- Natural — standard rhythm
- Brisk — quicker pace, shorter shots (morning storytime, car rides)

5. Core Features

5.1 Voice Capture

- Input: browser-based continuous audio capture via Web Audio API
- End-of-speech detection: 4-second silence trigger
- Max duration: 5 minutes per recording

5.2 Transcription and Cleanup

A silent preprocessing step. The user is never shown a distinction between “raw” and “cleaned” transcripts.

- Transcription: Whisper large-v3 with word-level timestamps
- Cleanup pass (single fast model call): removes long pauses, filler words (um, uh, like, you know), and obvious self-corrections where the user restated the same phrase. Preserves all story content, character names, and intentional repetition. Runs in under 2 seconds.
- Not included: deep narrative reconstruction, inference of missing plot points, or rewriting broken stories. The user is expected to tell a reasonably coherent story; cleanup removes artifacts only.

5.3 Scene Planning

- Model: K2 Think
- Input: cleaned transcript + selected filters
- Output (structured JSON): scene beats with duration, visual description, character references, setting, tone cue, and ambient tag
- Character sheet: the model emits a stable description of each recurring character so all downstream generations reference the same visual identity

5.4 Story Rewrite (Reading Level and Tone)

- Model: K2 Think, single call applying both reading-level and tone transforms
- Input: cleaned transcript + reading-level filter + tone filter
- Output: the final narration script — the actual words the narrator will speak

5.5 Visual Generation

- Primary: Imagen 3 via Google AI Studio free tier
- Fallback: FLUX.1 schnell via fal.ai signup credits
- Consistency: character reference image generated first, passed as reference to all subsequent scene generations
- Parallelization: all scene images generated concurrently

5.6 Narration Generation

- Source: Gemini 2.5's native multimodal audio generation — no external service
- Voice configuration: one of six preset voice profiles mapped from the Narrator Voice filter
- Pacing control: inserted pauses and speaking-rate parameters from the Pacing filter
- Rendering: one continuous narration pass over the full script so prosody flows naturally across sentences

5.7 Ambient Audio

- Source: Gemini multimodal audio (same stack as narration)
- Mix: approximately -22 dB below narration, atmospheric not distracting

5.8 Assembly

- Compositor: FFmpeg
- Timing anchor: scene durations derived from narration audio length so visuals accommodate voice, not vice versa
- Motion: Ken Burns pan-and-zoom per image, scaled to scene duration
- Transitions: 0.5-second crossfades
- Output: MP4 H.264 1080p

5.9 The Weaving Screen

- User's words appear as handwritten calligraphy at Whisper-timestamp rhythm — the original voice, made visible
- As the rewrite runs, words slowly dissolve and reform into the narrator's phrasing — the transformation is literal, not hidden
- Character sketches fade in on screen edges as the model identifies them
- Main canvas paints scenes with per-filter appearance animations
- Gold-circle timeline at the bottom fills as each scene completes
- No spinners, no percentage bars, no status text — latency becomes ritual

6. Technical Architecture

6.1 Stack

Layer	Technology	Cost
Frontend	React + Vite, Tailwind CSS	Free
Backend	Node.js + Express, FastAPI for ML calls	Free
Transcription	Whisper large-v3 on Modal or Replicate	~\$0.01 / story
Cleanup + planning + rewrite	K2 Think (primary), Gemini 2.5 Pro (fallback)	Free
Image generation	Imagen 3 via Google AI Studio (primary), FLUX.1 schnell via fal.ai (fallback)	Free tier
Narration audio	Gemini 2.5 native multimodal TTS	Free tier
Ambient audio	Gemini 2.5 multimodal audio	Free tier
Assembly	FFmpeg	Free
Storage	Supabase (session cache, temporary MP4 hosting)	Free tier
Deployment	Vercel (frontend), Modal or Fly.io (backend)	Free tier

6.2 Data Flow

- User records audio; audio streams to backend continuously
- Whisper transcribes with word-level timestamps
- Model runs the cleanup pass (removes pauses, fillers, obvious restarts)
- Model runs scene plan + reading-level rewrite + tone adjustment in a single call
- Character reference image generates first (2–4 seconds)
- Scene images generate in parallel using character reference (3–5 seconds)
- Gemini generates continuous narration audio + per-scene ambient audio in parallel
- FFmpeg assembles once all assets land, with scene durations anchored to narration timing
- Total wall-clock: 60–90 seconds

6.3 Filter Cache Strategy

Each filter axis maps to a specific pipeline stage. Changing one axis invalidates only caches downstream of that stage.

Axis Changed	Invalidated	Regeneration Time
Visual Style	Character reference + all scene images	~10 seconds
Narrator Voice	Narration audio only	~8 seconds
Reading Level	Rewrite + narration (images stay)	~15 seconds
Tone	Rewrite + narration + some image mood cues	~18 seconds

Axis Changed	Invalidated	Regeneration Time
Pacing	Narration timing + assembly	~6 seconds

7. Cost Model

Component	Per Story	Weekend Total
Whisper transcription	~\$0.01	Under \$1
K2 Think (cleanup + rewrite + planning)	Free	Free
Imagen 3 (free tier, 1500/day)	\$0.00	\$0.00
Gemini multimodal audio	Free tier	Free
Infrastructure	—	Free tier
Projected total	~\$0.01	Under \$5

8. 36-Hour Build Plan

Hours 0–6 — Foundation

- Project scaffolding: React frontend, Node backend, Supabase wiring
- Browser audio capture end-to-end
- Whisper integration returning transcript with word timestamps
- Cleanup call: raw transcript in, clean transcript out
- First Imagen call producing a usable image
- First Gemini multimodal narration call producing audio

Hours 6–14 — Core Pipeline

- Model scene planning emitting structured JSON with character references
- Parallelized image generation with character consistency
- Narration audio generation via Gemini multimodal
- FFmpeg assembly with narration-anchored timing
- Two visual styles end-to-end (Watercolor, Ghibli)
- End-to-end demo working with default filter settings

Hours 14–22 — The Filter Matrix

- Remaining four visual styles (Pixar, Paper Cutout, Charcoal, Crayon)
- All six narrator voices tested and tuned
- Reading-level rewrite: verify genuinely different outputs at all four levels on the same input
- Tone and pacing filters wired through model prompt and narration parameters
- Filter cache strategy implemented end-to-end

Hours 22–28 — Refilter Experience

- Picker UI with all five axes and pre-selected defaults
- Refilter flow: change axis → regenerate only affected layers → play updated film
- Adversarial testing of character consistency across 30+ test stories
- Adversarial testing of cleanup: 20+ transcripts with real storytelling artifacts

Hours 28–32 — The Weaving Screen

- Candle flame + silence-detection transition
- Word-by-word calligraphy render at Whisper-timestamp rhythm — then the word-reformation animation as rewrite runs
- Per-filter scene appearance animations
- Character sketch edge animations
- Gold-circle timeline

Hours 32–35 — Polish and Fallbacks

- Pre-generate all six filter preview tiles
- Pre-render two fallback films of the rehearsed demo story
- Typography pass: Georgia serif, warm amber, deep navy
- Full demo rehearsal

Hours 35–36 — Demo Day Setup

- Booth staging: dim lamp, headphones, printed cards, laptop power
- Network and backup-laptop checks
- Final script rehearsal with a teammate as judge

9. Demo Plan

Booth Setup

- One laptop facing judges
- Over-ear headphones on the table
- One dim desk lamp
- Printed 4×6 cards: “Tell a story. Any voice. Any age. Any world.”
- No posters, no stickers — the calm tone reads differently than other booths

The 2-Minute 30-Second Script

0:00–0:15 — Opening.

“Every night, millions of parents tell bedtime stories. The story is usually great. But kids ask for the same one a week later, and by then they’ve grown up a year. And here’s the thing: a story doesn’t have one correct form. A three-year-old and a ten-year-old can hear the same story — same plot, same fox, same star — each version true to what they need. Lullaby makes that literal.”

0:15–0:35 — Picker walkthrough.

Tap Begin. Picker appears. “Five dimensions — visual style, narrator voice, reading level, tone, pacing. Two thousand one hundred and sixty combinations.” Defaults are set; just tap Record.

0:35–1:05 — Recording.

Read the Juniper story at a natural, calm pace. Don’t perform — just tell it.

1:05–1:50 — Weaving screen.

Say nothing for 20 seconds; let the judges watch. Point quietly to the word transformation: “Watch the words change. That’s the story moving from my voice into the narrator’s.”

1:50–2:05 — Premiere.

Slide headphones across the table to the judge. They tap Play. Film plays — watercolor Juniper, grandmother narration, cozy and unhurried.

2:05–2:20 — Reading level swap.

“Same story, for a younger child.” Change Reading Level to Toddler. 15-second regeneration. Replay: same plot, very short sentences, sensory repetition. The judges hear the same story told at a completely different age.

2:20–2:35 — Voice swap.

“Same story, different voice.” Change Narrator Voice to Kid Narrator. 8-second regeneration. Replay: now a seven-year-old is narrating.

2:35 — Close.

“One story. Any voice. Any age. Any world. And some day, when that child is twenty-five, they’ll be able to play back the version their grandmother told, at the age they were when they first heard it. Every parent could use this tonight.”

The Rehearsed Story

“Once there was a little fox named Juniper who was afraid of the dark. One night, a star fell into her garden. It was small, and warm, and it whispered to her, ‘Come with me.’ Juniper took the star’s hand, and together they walked into the night, and she was not afraid anymore.”

30 seconds when read naturally. One character, one antagonist force, one setting, one magical element, one emotional arc. Every filter combination renders this beautifully.

Fallbacks

- Network failure: pre-rendered watercolor demo film plays from local disk
- Pipeline fails mid-demo: pre-rendered fallback of the Juniper story is cached; swap via feature flag
- Imagen rate-limited: auto-swap to fal.ai FLUX

- Gemini audio rate-limited: backup narration audio for the demo story pre-rendered and ready
- Laptop crash: identical backup laptop preloaded, reachable in under 30 seconds

10. Risks and Mitigations

Risk	Impact	Mitigation
Narration sounds robotic or wrong for the tone	Critical	Test all six voice profiles against all five tones and four reading levels; ship only combinations that consistently land
Gemini native TTS is inaccessible or inadequate	Critical	Confirm access hour 0; if inadequate, document the constraint change before proceeding
Character consistency breaks across scenes	High	Generate character reference first; use as reference for all scenes; test across 30+ stories
Cleanup misses pauses or fillers that end up in narration	Medium	Test against 20+ transcripts; add silence-trimming pass on Whisper timestamps as fallback
Reading-level rewrites feel indistinguishable	High	Explicit rewrite examples baked into model prompt; A/B test same input at all four levels before demo
Narration timing drifts from visual scene timing	High	Narration drives timing, visuals accommodate; scene durations computed from narration audio length
Pipeline latency over 90 seconds live	High	Weaving screen turns wait into ritual; aggressive parallelization; pre-rendered fallback always ready
Venue network unreliable	High	Mobile hotspot backup; pre-rendered demo on disk as last resort

11. Success Metrics

For the Hackathon

- End-to-end live demo produces a watchable narrated film from spoken recording in under 90 seconds
- At least three filter axes demonstrated (visual style, reading level, narrator voice)
- Refilter regenerates affected layers in under 30 seconds
- Character consistency maintained across 10+ scenes
- At least one judge audibly reacts during the demo
- Crowd forms at the booth during judging rounds

Beyond the Hackathon

- Percentage of sessions completing record → play loop
 - Percentage of users who swap at least one filter after first playback
 - Distribution of reading-level selections
 - Return sessions per household per week
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12. Future Scope

Explicitly out of scope for the hackathon. What follows is the long game.

The Memory Artifact

The most emotionally resonant direction is not a feature — it's a format. A family builds a canon over time: the Juniper stories, told across years, each one stored and indexed. The system retains not just the films but the metadata of the moment: who told the story, what voice was chosen, what reading level was right for the child then. When that child is twenty-five, they can play back the watercolor version their grandmother narrated when they were four. The story is a time capsule. Lullaby is the format it was preserved in.

Feature Directions

- Returning character memory: characters stay visually consistent across multiple stories over time — Juniper looks like Juniper next month
 - Custom narrator voices via opt-in voice sample — a grandmother's actual voice, preserved as a preset
 - Multilingual storytelling: narration in a different language than the input
 - A sixth axis — music — with generated soundtrack beds tuned to tone
 - Printed storybook export: generated images + narration script as a PDF picture book
 - Co-creation with the child: kid draws a character, camera captures it, AI uses it as the protagonist
 - Family library: all stories in one place, browseable by date, character, reading level, and voice
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13. Team and Attribution

Lead: Sayan Gupta — architecture, pipeline prompt engineering, product direction

Collaborators: TBD

Services and Credits

- K2 Think — transcript cleanup, scene planning, reading-level and tone rewrites
- OpenAI Whisper — speech-to-text (self-hosted)

- Google Gemini 2.5 — multimodal audio generation for narration and ambient
- Google Imagen 3 via AI Studio — primary image generation
- Black Forest Labs FLUX.1 via fal.ai — fallback image generation
- FFmpeg — video assembly
- Supabase, Vercel, Modal — infrastructure

Tell a story. Any voice. Any age. Any world.